HOST SUSCEPTIBILITY, IMMUNITY AND VACCINATION

Prof. Dr. Fouad Metry Atia

Professor of community medicine , faculty of medicine

INTENDED LEARNING OUTCOME

- **By the end of the lecture the student will be able to:**
  - **Define immunity and identify the types of immunity**
  - **Differentiate between active and passive immunity**
  - **Describe the host defense mechanisms**
  - **Identify the different types of vaccines**
  - **Define Sero prophylaxis and its uses**
  - **Differentiate between chemo and Sero prophylaxis**

RESISTANCE (NONSPECIFIC DEFENSE MECHANISMS)

- **Definition: Natural body mechanisms that block invasion of infectious agents.**
- **Key Features:**
  - **Not dependent on antibodies.**
  - **Not related to prior infection or immunization.**
  - **Based on species, race, sex, personal habits**

NONSPECIFIC DEFENSE MECHANISMS (RESISTANCE):

First line of defense	Second line of defense
• Skin • Mucous membranes • Secretions of skin and mucous Membranes	• Phagocytic White blood cells • Antimicrobial proteins •The inflammatory response

IMMUNITY

- Derived from Latin immunis = exemption
- It is the ability of the body to recognize and defend itself against specific pathogens or antigen
- Represents specific resistance (third line of defense

1. ACTIVE IMMUNITY:

- This develops only **after exposure** to inducing agents such as microbes, toxins, or other foreign substances.
- It occurs when a person is exposed to a live pathogen, and develops a primary immune response, which leads to immunological **memory**.
- This type of immunity is "natural" because it is **not induced by man**.

- **The degree of post infection immunity varies and it may be:**
  - **Absolute as in yellow fever where the level of immunoglobulin had never declined after contracting clinical attack of yellow fever and recovery.**
  - **Solid (lifelong immunity) as in the case of small pox, chicken pox, measles, German measles, mumps.**
  - **Moderate (lasting for few years) as in the case of typhoid fever, M. meningitis and pertussis or**
  - **Transient (lasting a short period of time) as in common cold and influenza.**

ARTIFICIALLY ACQUIRED ACTIVE IMMUNITY

- It can be induced by a vaccine, a substance that contains antigen.
- A vaccine stimulates a primary response against the antigen without causing symptoms of the disease.

PASSIVE IMMUNITY

- It is the transfer of active immunity, in the form of ready-made antibodies, from one individual to another.
  - Naturally acquired passive immunity : Passive immunity can occur naturally, when maternal antibodies are transferred to the fetus through the placenta or through breast milk .
  - Artificially acquired passive immunity : This is a short-term immunization induced by the transfer of antibodies, which can be administered in several forms; as human or animal plasma or serum, as pooled human immunoglobulin

IMMUNIZATION

- It is a procedure in which **immunobiological substances** are administered to strengthen the defense (immune) mechanism as to protect the individual against the disease.
- When this is implemented in the community at large, the **susceptible** population can be reduced.
- There are two types of immunization—active and passive.

THE DIFFERENCES BETWEEN ACTIVE AND PASSIVE IMMUNIZATION

Feature	Active Immunization	Passive Immunization
Substance	Antigens	Antibodies
Onset	10–15 days	Immediate
Process	Active	Passive
Duration	Long-lasting	Short-lived
Reactions	Rare	Common
Timing	Before exposure	After exposure
Treatment use	Not used	Often used (e.g., tetanus, diphtheria)
Utility	More useful	Less useful

TYPES OF VACCINE

This include

- **Live vaccine**
- **Live (attenuated)**
- **Inactivated (killed)**
- **Toxoids**
- **Subunit**
- **Recombinant**
- **Combined vaccines**
- **Tissue culture vaccines**

TYPES OF VACCINE

- **Live vaccines**
  - **Contain live vaccinia virus (nonpathogenic but antigenic)**
  - **Example: Smallpox vaccine**
- **Live attenuated vaccines**
  - **Prepared from weakened organisms (via subculture/unfavorable conditions)**
  - **Examples: BCG, Oral Typhoid, Measles, Mumps, Rubella, Oral Polio (Sabin), Yellow fever (17D), Influenza, Epidemic typhus**
- **Inactivated (Killed) vaccines**
  - **Organisms killed by heat/chemicals; safe but less efficacious**
  - **Require multiple doses + boosters**
  - **Examples: Typhoid, Pertussis, Cholera, Plague, Salk polio, Sinopharm (COVID-19)**

- **Toxoids**
  - Detoxicated toxins; induce antibodies against toxin, not infection
  - Very safe and effective
  - Examples: Diphtheria, Tetanus
- **Cellular fraction vaccines**
  - Prepared from bacterial cell wall/capsule extracts
  - Safe, effective, some need boosters
  - Examples: Meningococcal, Pneumococcal, Haemophilus influenzae
- **Subunit vaccines**
  - Made from a component of the virus
  - Example: Influenza vaccine

- **Recombinant vaccines**
  - **Genetically engineered using recombinant DNA technology**
  - **Example: Hepatitis B vaccine**
- **Combined (mixed) vaccines**
  - **Contain more than one antigen**
  - **Examples: Pentavalent (DPT + HBV + Hib), DPT, DT, MMR**
- **Tissue culture vaccines**
  - **Seed viruses cultured in special cells (chick embryo, vero cells, fibroblasts)**
  - **Example: HDCV (Human diploid cell rabies vaccine)**

CONTRAINDICATIONS OF VACCINATION

- Fever, mild respiratory infection and other minor illness are not Contra indications to vaccination.
- Presence of malnutrition is also not a contraindication.
- Diarrhea is not a contraindication for OPV immunization, however the dose of OPV given during episodes of diarrhea should not be counted and another dose given at the earliest opportunity.

- **Breast-feeding prior to or after OPV administration is not forbidden nor contraindication. Only avoid giving hot liquid or food for ½ hour after OPV**
- **Tuberculin testing is not required prior to administration of BCG till the age of 12 years.**
- **Listening too many contraindications may leave many children unvaccinated and if ill or malnourished, they are more likely to die from a vaccine preventable disease.**

STRICT CONTRAINDICATION INCLUDE:

- Very ill children who need hospital admission.
- BCG is contraindication in AIDS and immune deficiency diseases.
- Pertussis component of DPT is contraindicated under the following conditions.
  - Children with neurological disorders.
  - Severe reactions to previous DPT dose.
  - Children over the age of five years.

CAUSES OF VACCINATION FAILURE (WHAT DAMAGE VACCINE)

- **Heat: Polio, measles, MMR, vaccines.**
- **Sunlight: BCG, DPT.**
- **Chemicals:**
- **Disinfecting and antiseptics, including spirits, are damaging to vaccines.**

  The site of injection should be left for a while after sterilization with alcohol to dry before injecting the vaccine.
- **Time: All vaccines lose their potency after certain time. This is recorded on the vaccine as expiry date.**
- **Reconstitution: Reconstituted vaccines like (BCG, measles, MMR) are damaged after few hours from reconstitution.**

SEROPROPHYLAXIS OR PASSIVELY ACQUIRED ARTIFICIAL IMMUNITY

- **ANTI-SERA**
  - **Prepared from plasma of immunized animals (e.g., horses)**
  - **Cheap, less effective, immunity lasts 2–3 weeks**
  - **Frequent reactions due to animal protein → test dose required**
  - **Examples: Anti-tetanus serum (ATS), Anti-diphtheria serum (ADS), Anti-snake venom, Anti-gas gangrene serumvenom; Anti-gas-gangrene serum.**

IMMUNOGLOBULINS

- Ready-made antibody preparations from human plasma
- Provide instant immunity → used for passive immunization
- Very safe (no reactions)
- Examples,
  - human rabies immunoglobulin (HRIg);
  - human tetanus immunoglobulin (HTIg);
  - hepatitis B immunoglobulin (HBIg);
  - varicella zoster immunoglobulin (VZIg).

CHEMOPROPHYLAXIS

- Preventive use of chemical substances/antibiotics to protect susceptibles
- Examples:
  - Antibiotics before surgery → prevent wound infections
  - INH for children exposed to pulmonary TB cases
  - Chloroquine for malaria prevention (except resistant P. falciparum)
  - Tetracycline for cholera prevention
  - Rifampicin for meningitis prevention

VACCINATION SCHEDULE OF THE EGYPTIAN CHILD DURING THE FIRST TWO YEARS OF LIFE

Age	Vaccine	Route	Dose
At birth	BCG	ID	0.05 ml
	OPV (Sabin, zero dose)	Oral	2 drops
	Hepatitis B (HBV, birth dose within 24 hrs)	IM	0.5 ml
2 months	OPV	Oral	2 drops
	Pentavalent (DPT + HBV + Hib)	IM	0.5 ml
	IPV (Salk)	IM	0.5 ml
4 months	OPV	Oral	2 drops
	Pentavalent (DPT + HBV + Hib)	IM	0.5 ml
	IPV	IM	0.5 ml
6 months	OPV	Oral	2 drops
	Pentavalent (DPT + HBV + Hib)	IM	0.5 ml
	IPV	IM	0.5 ml
9 months	Vitamin A	Oral	100,000 IU
12 months	OPV	Oral	2 drops
	MMR (Measles, Mumps, Rubella)	SC	0.5 ml
18 months	OPV	Oral	2 drops
	DPT booster	IM	0.5 ml
	MMR booster	SC	0.5 ml
	Vitamin A	Oral	200,000 IU

Thank you